Godlog - May 10, 2026
Single-Detector LLR/FAR Proposal for SPIIR

Local implementation plan for the Eric branch

SPIIR workflow notes

May 10, 2026

May 10, 2026 – Purpose of This Godlog

This file records the proposed single-detector implementation for the Eric branch. The intent is:

- use the main SPIIR pipeline as the stable upstream coherent search;
- use the `wguo-single-detector` branch as the baseline for single-detector row extraction/routing;
- add the missing piece: a single-detector likelihood-ratio rank and FAR assignment path;
- keep the proposal focused on the single branch only.

Key point

The final single-detector target is not a raw (ρ_m, χ_r^2) plot. The target calibrated product is

$$(\rho_m, -\log_{10} \mathrm{FAR}_{\mathrm{sd}}).$$

May 10, 2026 – Code Bases to Combine in the Eric Branch

Source	Role in the realization
Main SPIIR	Provides the online/offline postcoh pipeline, detector conditioning, SPIIR filtering, `cuda_postcoh`, coherent background/FAR, and `FinalSink`.
`wguo-single-detector`	Provides the single-detector extraction idea: preserve detector-local information from postcoh/coexistence rows so H1 and L1 can be analyzed as independent single-detector categories.
Eric branch	Should combine the two bases and add the new single-detector LLR/FAR layer described here.

`https://git.ligo.org/lscsoft/spiir`

`https://git.ligo.org/lscsoft/spiir/-/tree/wguo-single-detector?ref_type=heads`

The local code in this workspace is not a Git checkout, so the changes are prepared locally for later push by the user.

May 10, 2026 – What Already Exists

Upstream products already available

- detector-local complex SNR streams;
- local peak SNR amplitudes `snglsnr[j]`;
- detector-local autochisq values `chisq[j]`;
- row flags: foreground, background, empty;
- bank/template/time metadata;
- coherent quantities such as `cohsnr`, `nullsnr`, and `cmbchisq`.

Single-side interpretation

$$\rho_m = \texttt{snglsnr[j]}, \qquad \chi_r^2 = \texttt{chisq[j]}.$$

For each detector $j \in \{\mathrm{H1}, \mathrm{L1}\}$, one postcoh row can produce a detector-local feature

$$d_{j,m} = (\rho_m, \chi_r^2).$$

May 10, 2026 – What Is Missing

The current code can expose the needed observables, but the single branch still needs a complete statistical significance layer:

1. define H_0 and H_1 likelihoods for (ρ_m, χ^2_r);
2. compute a log-likelihood-ratio rank $r(d)$;
3. accumulate a single-detector rank background;
4. persist that background in a dedicated single FAR–LLR file;
5. assign FAR from the rank-tail count and livetime;
6. output $(\rho_m, -\log_{10} \mathrm{FAR}_{\mathrm{sd}})$ rows.

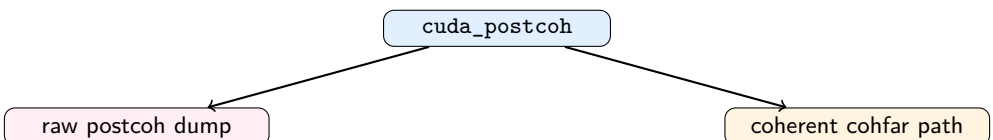
The single-detector background file is only a FAR–LLR/rank-livetime file. It is not the coherent XML statistics file and it is not a persistent two-dimensional (ρ_m, χ^2_r) map.

May 10, 2026 – Single-Branch Boundary in the Pipeline

Code anchor: `spiir/gstlal-spiir/python/pipemodules/spiirparts.py`; `mkPostcohSPIIROnline(...)`, after `mkcudapostcoh(...)`

The natural branch point is after `cuda_postcoh`, because this is the first point where every postcoh row already contains:

`snglsnr[j]`, `chisq[j]`, `is_background`, `ifos`, `tmplt_idx`, `bankid`.



The single branch must not recompute matched filtering, $C_{j,m}$, `cohsnr`, or `cmbchisq`. It reuses the detector-local pair already created upstream.

May 10, 2026 – Current Local Hook in `pipeline.sh`

Code anchor: `spiir/pipeline.sh`; mode parser and post-graph single analyzer call

```
PIPELINE_MODE="${PIPELINE_MODE:-}"
...
export PIPELINE_MODE
"${base_cmd[@]}"

single_far_csv="${jobno}/${jobno}_single_detector_far.csv"
single_far_background_input="${SINGLE_FAR_BACKGROUND_INPUT:-}"
single_far_background_output="${SINGLE_FAR_BACKGROUND_OUTPUT:-${jobno}/${jobno}_single_far_llr_background.json}"
```

In `-single` mode, the coherent GStreamer graph still runs. After that graph exits, the offline single analyzer reads `sdpostcoh*.xml.gz` and writes single-detector FAR rows.

May 10, 2026 – Local Code Added to Launch LLR/FAR

Code anchor: spiir/pipeline.sh; single_detector_cmd

```
single_detector_cmd=(
  python "${single_detector_py}" single
  --postcoh-glob "${jobno}/sdpostcoh*.xml.gz"
  --output "${single_far_csv}"
  --ifos H1,L1
  --min-snr 4
  --background-output "${single_far_background_output}"
  --snr-bins "${SINGLE_FAR_SNR_BINS:-4,6,8,12,inf}"
)
if [[ -n "${single_far_background_input}" ]]; then
  single_detector_cmd+=(--background-input "${single_far_background_input}")
fi
if [[ "${single_far_calibrate}" == "1" ]]; then
  single_detector_cmd+=(--calibrate-noise-dof)
fi
"${single_detector_cmd[@]}"
```

Cold start: if no `SINGLE_FAR_BACKGROUND_INPUT` is supplied, the run calibrates from the current background rows and writes the first `single_far_llr_background.json`. Later runs load that file and update it.

May 10, 2026 – Single-Detector Categories

The single branch only uses detector-local H1 and L1 information:

$$j \in \{\mathrm{H1}, \mathrm{L1}\}.$$

They are analyzed separately, not as a coherent HL network:

$$\mathrm{H1} \to \mathrm{H1\_sd}, \qquad \mathrm{L1} \to \mathrm{L1\_sd}.$$

For each available detector,

$$\rho_m = \texttt{snglsnr[j]}, \qquad \chi_r^2 = \texttt{chisq[j]}.$$

Key point

If only H1 or only L1 is available because the other detector loses lock, the available detector can still contribute to its own single-detector category.

May 10, 2026 – Coexistence Rows and the wguo Extraction Idea

A multi-detector coexistence row, for example HL or HLV, can still carry detector-local fields:

$$\texttt{snglsnr[H1]}, \quad \texttt{chisq[H1]}, \quad \texttt{snglsnr[L1]}, \quad \texttt{chisq[L1]}.$$

The single branch should copy the detector-local H1/L1 pairs into the single categories:

$$\mathrm{H1} : (\texttt{snglsnr[H1]}, \texttt{chisq[H1]}) \to \mathrm{H1\_sd},$$

$$\mathrm{L1} : (\texttt{snglsnr[L1]}, \texttt{chisq[L1]}) \to \mathrm{L1\_sd}.$$

This is not numerical subtraction. The coherent row remains in the coherent branch; the single branch only duplicates the local H/L pair for independent single-detector ranking.

May 10, 2026 – Feature Extraction From Postcoh Rows

Code anchor: `spiir/gstlal-spiir/python/pipemodules/single_detector_far.py`; `features_from_postcoh_row`

For every postcoh row and every requested detector j, the analyzer reads:

$$\rho_m = \texttt{snglsnr[j]}, \qquad \chi_r^2 = \texttt{chisq[j]}.$$

It rejects missing, nonpositive, or below-threshold values. The retained object is:

$$d_{j,m} = (\rho_m, \chi_r^2).$$

The single branch does not inspect the full SNR stream at this stage. It trusts `cuda_postcoh` to have already selected the trigger peak and computed the detector-local autochisq around that peak.

May 10, 2026 – Meaning of ρ_m

For fixed detector j and template m, SPIIR has a complex SNR time series:

$$\rho_{j,m}(t).$$

A trigger is centered at one detector-local peak time t_j. The complex peak value is

$$\hat{\rho}_{j,m} = \rho_{j,m}(t_j).$$

The single-detector SNR coordinate is the peak amplitude:

$$\rho_m = |\hat{\rho}_{j,m}| = \texttt{snglsnr[j]}.$$

Key point

ρ_m is the peak SNR for this detector and this template. It is not `cohsnr`, because no network projection is used in the single branch.

May 10, 2026 – Meaning of χ_r^2

The detector-local shape statistic compares the SNR samples around the peak to the stored local template shape:

$$r_{j,m}(\Delta) = \rho_{j,m}(t_j + \Delta) - \hat{\rho}_{j,m} C_{j,m}(\Delta), \qquad \Delta = -L, \ldots, L.$$

The reduced statistic is

$$\chi_r^2 \equiv \chi_{j,m}^2 = \frac{1}{N_{j,m}} \sum_{\Delta=-L}^{L} |r_{j,m}(\Delta)|^2.$$

In code this is stored as:

$$\chi_r^2 = \texttt{chisq[j]}.$$

Small χ_r^2 means the local window is consistent with the template shape; large χ_r^2 is more glitch-like.

May 10, 2026 – Likelihood Hypotheses

The new rank uses two statistical hypotheses:

- H_0 noise/glitch hypothesis: the local peak is noise-like. The residual shape has no deterministic signal-like shift.
- H_1 signal hypothesis: the local peak is signal-like. The residual may contain Gaussian noise plus a small deterministic template-mismatch shift.

The desired rank is a log-likelihood ratio:

$$r(d) \simeq \ln \frac{P(d \mid H_1)}{P(d \mid H_0)}, \qquad d = (\rho_m, \chi_r^2).$$

Using the product rule:

$$r(d) = \ln \frac{P(\chi_r^2 \mid \rho_m, H_1)}{P(\chi_r^2 \mid \rho_m, H_0)} + \ln \frac{P(\rho_m \mid H_1)}{P(\rho_m \mid H_0)} + \text{const.}$$

May 10, 2026 – Why the Noise Side Is Chi-Square-Like

In the ideal exact-match case,

$$\rho_{j,m}(t_j + \Delta) = \hat{\rho}_{j,m} C_{j,m}(\Delta) + n(\Delta),$$

so the residual is

$$r_{j,m}(\Delta) = n(\Delta).$$

After whitening, the local noise is modeled as Gaussian in the complex components. Therefore

$$|r_{j,m}(\Delta)|^2 = r_{\mathrm{re}}(\Delta)^2 + r_{\mathrm{im}}(\Delta)^2$$

is a sum of Gaussian squares, and the local quadratic power is chi-square-like.

The actual online trigger stream is peak-selected and clustered, so the ideal degree count is not trusted directly. We use an empirical effective degree of freedom ν_{eff}.

May 10, 2026 – Noise-Side Likelihood

Under H_0, there is no deterministic mismatch shift:

$$\lambda = 0.$$

The working model is a reduced central chi-square:

$$Y = \nu_{\mathrm{eff}} \chi_r^2, \qquad Y \mid H_0 \sim \chi^2_{\nu_{\mathrm{eff}}}.$$

Therefore the density for the observed reduced statistic is

$$P(\chi_r^2 \mid \rho_m, H_0) = \nu_{\mathrm{eff}} f_{\chi^2_{\nu_{\mathrm{eff}}}}\left(\nu_{\mathrm{eff}} \chi_r^2\right).$$

Key point

ν_{eff} is not calculated from a single trigger. It must be calibrated from many background/noise triggers in the same category, optionally in SNR bins.

May 10, 2026 – Calibrating ν_{eff}

For a background category, such as $\mathrm{H1\_sd}$, collect background triggers and read:

$$\chi^2_{r,k} = \texttt{chisq[j]}, \qquad k = 1, \ldots, N_{\mathrm{bg}}.$$

If the reduced statistic follows

$$\chi^2_r \sim \frac{1}{\nu_{\mathrm{eff}}} \chi^2_{\nu_{\mathrm{eff}}},$$

then

$$\mathrm{Var}(\chi^2_r) = \frac{2}{\nu_{\mathrm{eff}}}.$$

So a simple calibration is

$$\nu_{\mathrm{eff}} \approx \frac{2}{\mathrm{Var}_{\mathrm{bg}}(\chi^2_r)}.$$

The local code now supports SNR-bin calibration:

$$\rho_m \in [4, 6), [6, 8), [8, 12), [12, \infty).$$

May 10, 2026 – Signal-Side Mismatch Model

Under H_1, a real signal may not perfectly match the stored local template shape. Write

$$s_{j,m}(\Delta) = \hat{\rho}_{j,m} C_{j,m}(\Delta)\big(1 + \beta(\Delta)\big).$$

Then the tested residual becomes

$$r_{j,m}(\Delta) = n(\Delta) + \beta(\Delta)\hat{\rho}_{j,m} C_{j,m}(\Delta).$$

Within the short local window, approximate

$$\beta(\Delta) \approx \bar{\beta}.$$

Key point

The residual is Gaussian noise plus a deterministic mismatch shift. This is why the signal-side shape likelihood is modeled with a noncentral chi-square distribution.

May 10, 2026 – The Noncentrality Parameter λ

Define the deterministic mismatch shift

$$\mu(\Delta) = \bar{\beta}\,\hat{\rho}_{j,m} C_{j,m}(\Delta).$$

For a sum of squared Gaussian variables with deterministic mean shift, the noncentrality is the squared norm of that shift:

$$\lambda = \sum_{\Delta=-L}^{L} |\mu(\Delta)|^2.$$

Substituting $\mu(\Delta)$ gives

$$\lambda(\bar{\beta}, \rho_m) = \bar{\beta}^2 \rho_m^2 \sum_{\Delta=-L}^{L} |C_{j,m}(\Delta)|^2.$$

λ is not a new measured feature. It is the noncentral chi-square parameter introduced by the likelihood model. It grows with mismatch level, peak SNR, and local template-shape power.

May 10, 2026 – Signal-Side Likelihood

For fixed $\bar{\beta}$, use

$$Y = \nu_{\text{eff}} \chi_r^2, \qquad Y \mid (\rho_m, \bar{\beta}, H_1) \sim \chi^2_{\nu_{\text{eff}}}(\lambda).$$

Thus

$$P(\chi_r^2 \mid \rho_m, \bar{\beta}, H_1) = \nu_{\text{eff}} f_{\chi^2_{\nu_{\text{eff}}}(\lambda)}\left(\nu_{\text{eff}} \chi_r^2\right).$$

Because the true $\bar{\beta}$ is unknown event by event, treat it as a nuisance parameter:

$$\bar{\beta} \sim \text{Uniform}(0, \beta_{\text{max}}).$$

May 10, 2026 – Marginalized Signal Likelihood

The final signal-side likelihood averages over allowed mismatch levels:

$$P(\chi_r^2 \mid \rho_m, H_1) = \frac{1}{\beta_{\max}} \int_0^{\beta_{\max}} \nu_{\mathrm{eff}} f_{\chi^2_{\nu_{\mathrm{eff}}}(\lambda(\bar{\beta},\rho_m))}\left(\nu_{\mathrm{eff}}\chi_r^2\right) \mathrm{d}\bar{\beta}.$$

In the code this is evaluated by a discrete beta grid:

$$P(\chi_r^2 \mid \rho_m, H_1) \approx \sum_i w_i \, g_i(\chi_r^2; \rho_m, \beta_i), \qquad \sum_i w_i = 1.$$

The prototype default uses $\beta_{\max} = 0.03$ and a uniform beta grid. The grid and weights are implementation choices that should later be calibrated or validated with injections/background.

May 10, 2026 – Full Single-Detector Rank

The full single-detector rank is

$$r(d) = \ln \frac{P(\chi_r^2 \mid \rho_m, H_1)}{P(\chi_r^2 \mid \rho_m, H_0)} + \frac{1}{2}\rho_m^2 + \text{const.}$$

The term $+\rho_m^2/2$ is the SNR-amplitude contribution from the matched-filter likelihood after maximizing over amplitude/phase/time.

For ranking, any fixed prior odds only add a constant:

$$\ln \frac{P(H_1 \mid d)}{P(H_0 \mid d)} = r(d) + \ln \frac{P(H_1)}{P(H_0)}.$$

Therefore $r(d)$ is enough to order events.

May 10, 2026 – Local Likelihood Implementation

Code anchor: spiir/gstlal-spiir/python/pipemodules/single_detector_far.py; SingleDetectorLikelihoodModel

```
def log_signal_shape_pdf(self, rho, chisq, autocorr_power=None, ifo=None):
    dof = self.signal_dof_for(rho, ifo)
    x = dof * float(chisq)
    terms = []
    for beta, weight in zip(self.beta_grid, self.beta_weights):
        lam = self.noncentrality(rho, beta, autocorr_power)
        terms.append(math.log(weight) + noncentral_chisq_logpdf(
            x, dof, lam, max_terms=self.ncx2_max_terms))
    return math.log(dof) + logsumexp(terms)

def log_noise_shape_pdf(self, rho, chisq, ifo=None):
    dof = self.noise_dof_for(rho, ifo)
    x = dof * float(chisq)
    return math.log(dof) + central_chisq_logpdf(x, dof)
```

The factor $\log(\nu_{\mathrm{eff}})$ is the Jacobian from $Y = \nu_{\mathrm{eff}}\chi^2_r$.

May 10, 2026 – Local Rank Implementation

Code anchor: `spiir/gstlal-spiir/python/pipemodules/single_detector_far.py`; `log_likelihood_ratio and rank`

```
def log_likelihood_ratio(self, rho, chisq, autocorr_power=None, ifo=None):
    shape_llr = (
        self.log_signal_shape_pdf(rho, chisq, autocorr_power, ifo=ifo)
        - self.log_noise_shape_pdf(rho, chisq, ifo=ifo))
    snr_llr = self.snr_log_weight * float(rho) * float(rho)
    return shape_llr + snr_llr + self.rank_offset

def rank(self, rho, chisq, autocorr_power=None, ifo=None):
    return self.log_likelihood_ratio(rho, chisq, autocorr_power, ifo=ifo)
```

This turns every single-detector feature $d = (\rho_m, \chi_r^2)$ into one scalar r. Larger r means more signal-like.

May 10, 2026 – Rank-Space Background

For every single-detector background trigger, compute

$$r_k^{(\mathrm{bg})} = r(d_k^{(\mathrm{bg})}).$$

For a foreground candidate with rank r_*, count same-category background triggers with rank at least as large:

$$N_{\mathrm{bg}}^{(j)}(\geq r_*) = \sum_{k=1}^{N_{\mathrm{bg}}^{(j)}} \mathbf{1}\left(r_k^{(\mathrm{bg},j)} \geq r_*\right).$$

The indicator function is one if the condition is true and zero otherwise.

Key point

This is the single-detector replacement for the old two-dimensional coherent rank-map lookup. The significance calculation is one-dimensional in r.

May 10, 2026 – FAR From the Rank Tail

Let $T_{\mathrm{bg}}^{(j)}$ be the background livetime for the same detector category. The final single-detector FAR is

$$\mathrm{FAR}_{\mathrm{sd}}^{(j)}(r_*) = \frac{N_{\mathrm{bg}}^{(j)}(\geq r_*)}{T_{\mathrm{bg}}^{(j)}}.$$

The final plotted coordinate is

$$\left(\rho_m, -\log_{10} \mathrm{FAR}_{\mathrm{sd}}^{(j)}\right).$$

H1 and L1 must keep separate background rank lists and separate livetimes before any final display/alert-level merge.

May 10, 2026 – Local Background Accumulator

Code anchor: `spiir/gstlal-spiir/python/pipemodules/single_detector_far.py; RankBackground`

```
class RankBackground(object):
    def __init__(self):
        self._ranks = []
        self.livetime = 0.0

    def add_rank(self, rank):
        bisect.insort(self._ranks, float(rank))

    def count_ge(self, rank):
        idx = bisect.bisect_left(self._ranks, float(rank))
        return len(self._ranks) - idx

    def far(self, rank):
        if self.livetime <= 0.0:
            return 0.0
        return self.count_ge(rank) / self.livetime
```

This exact sorted-list version is suitable for offline validation. A production version can replace it with a compact histogram or quantile/rank summary if memory becomes too large.

May 10, 2026 – Single FAR–LLR Background File

Code anchor: `spiir/gstlal-spiir/python/pipemodules/single_detector_far.py`; `SingleFarLlrBackgroundFile`

The single background file stores only what FAR assignment needs:

- detector categories, e.g. $\mathrm{H1\_sd}$, $\mathrm{L1\_sd}$;
- background LLR/rank values $r_k^{(\mathrm{bg},j)}$;
- category livetimes $T_{\mathrm{bg}}^{(j)}$;
- the likelihood/calibration model used to compute those ranks.

$$\mathcal{B}_j = \left\{ r_1^{(\mathrm{bg},j)}, \ldots, r_N^{(\mathrm{bg},j)}; T_{\mathrm{bg}}^{(j)}; \nu_{\mathrm{eff}}(\rho_m) \right\}.$$

Key point

This file is the single-detector FAR–LLR file. It is not a coherent-style `cohfar` XML statistics file.

May 10, 2026 – Cold Start and Real-Time Update Logic

At the beginning, no single FAR–LLR background file exists. The first stage is a bootstrap:

1. run the single analyzer over about one week of data;
2. collect background rows and empty-row livetime;
3. calibrate ν_{eff} from background χ^2_r;
4. compute ranks for background and foreground rows;
5. assign provisional FAR from the in-run background;
6. write the first single FAR–LLR background file.

After that:

1. load the existing single FAR–LLR background file;
2. assign FAR to new foreground candidates using the accumulated background;
3. insert new background ranks and livetime;
4. write an updated background file snapshot.

May 10, 2026 – Persistent Background JSON

Code anchor: `spiir/gstlal-spiir/python/pipemodules/single_detector_far.py; SingleFarLlrBackgroundFile.dump`

```
data = {
    "version": self.VERSION,
    "created_utc": datetime.datetime.utcnow().isoformat() + "Z",
    "description": "single-detector FAR-LLR rank background and livetime",
    "ifos": list(self.ifos),
    "likelihood_model": self.model.to_dict(),
    "backgrounds": dict(
        (ifo, bg.to_dict()) for ifo, bg in self.backgrounds.items()),
}
```

This makes the file self-consistent: the same likelihood calibration used to create the old ranks can be reloaded before assigning FAR in later runs.

May 10, 2026 – Two Background Layers

The proposal separates two related but different background tasks:

Feature-space calibration

Use background (ρ_m, χ_r^2) points to fit or validate model parameters:

$$\nu_{\text{eff}}(\rho_m), \quad \beta_{\max}, \quad \text{tail behavior.}$$

This is for likelihood calibration, not direct FAR lookup.

Rank-space FAR assignment

Map every background trigger to r, then assign FAR from the one-dimensional rank tail:

$$(\rho_m, \chi_r^2) \to r \to \frac{N_{\text{bg}}(\geq r_*)}{T_{\text{bg}}}.$$

May 10, 2026 – How Foreground, Background, and Empty Rows Are Used

Row type	Single-branch action
`FOREGROUND`	Extract H1/L1 local features, compute r_*, assign FAR after the background state is available, and write output rows.
`BACKGROUND`	Extract H1/L1 local features, compute r_{bg}, and insert into the detector/category rank background.
`EMPTY`	Increment category livetime $T_{\mathrm{bg}}^{(j)}$. These rows are essential because FAR is a rate, not just a probability.

The local code now collects foreground features while accumulating background and livetime, then assigns FAR after the background stage.

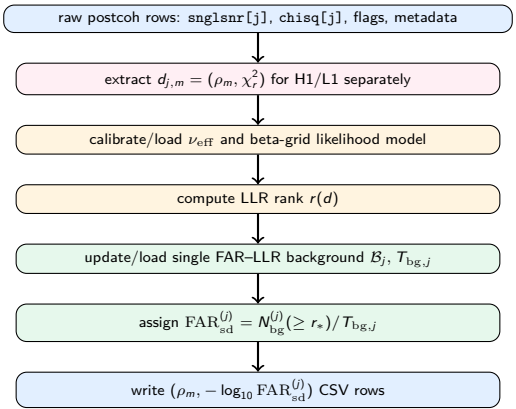
May 10, 2026 – Implementation Files Modified Locally

File	Local change
`spiir/pipeline.sh`	Adds single FAR–LLR input/output environment variables, cold-start calibration switch, and passes background options to the single analyzer.
`spiir/gstlal-spiir/python/pipemodules/single_detector_far.py`	Adds persistent background file IO, ν_{eff} calibration, reduced chi-square likelihood scaling, beta marginalization controls, rank background, and FAR assignment.
`spiir/gstlal-spiir/python/pipemodules/Makefile.am`	Installs `single_detector_far.py` and `combine_background_far.py` with the other pipemodules.

Key point

These changes are local. The user can push them to the GitLab Eric branch later.

May 10, 2026 – Expected Single-Branch Data Flow



May 10, 2026 – Validation Plan

The single branch needs validation in increasing order of risk:

1. **Syntax and import checks**: `python -m py_compile` for the new Python files; `bash -n pipeline.sh`.
2. **Unit checks**: central chi-square logpdf, noncentral Poisson-mixture logpdf, beta-mixture normalization, monotonic rank-tail count.
3. **Toy rows**: fake postcoh rows with known foreground/background/empty behavior.
4. **Small OzSTAR segment**: run `bash pipeline.sh -single` on a short known-good frame/cache segment.
5. **Cold-start week**: build the first `single_far_llr_background.json`.
6. **History-file run**: rerun with `SINGLE_FAR_BACKGROUND_INPUT` and confirm FAR assignment uses loaded background while appending new ranks/livetime.

May 10, 2026 – Scientific Checks Before Trusting FAR

- Check that the background χ^2_r distribution is close enough to the fitted reduced chi-square model in each SNR bin.
- Check whether ν_{eff} depends strongly on detector, bank, template duration, or SNR.
- Validate β_{max} and the beta prior with injections; the current uniform prior is a modeling convention.
- Verify that H1 and L1 single categories have independent livetime accounting.
- Confirm that copied H/L detector-local fields from coexistence rows are not removed from the coherent branch.
- Decide how to handle zero exceedance counts: empirical upper limit $1/T_{\mathrm{bg}}$ or a fitted tail extrapolation.

May 10, 2026 – Main Risk: Rank Consistency Across Background Files

If an old background file contains ranks computed with one likelihood calibration, and the new run changes ν_{eff}, beta grid, or SNR weight, then old and new ranks are not directly comparable.

The local implementation handles this by storing the likelihood model in the FAR–LLR background JSON.

Recommended production policy:

1. if `SINGLE_FAR_BACKGROUND_INPUT` is used, load its likelihood model and do not recalibrate by default;
2. if a new calibration is required, rebuild the rank background from raw stored background features or start a new background file version;
3. store model version and calibration metadata in every background file.

May 10, 2026 – Minimal Completion Sequence

1. Merge main SPIIR and `wguo-single-detector` extraction into Eric branch.
2. Ensure `pipeline.sh -single` creates `sdpostcoh*.xml.gz` while coherent branch continues.
3. Install `single_detector_far.py` and `combine_background_far.py`.
4. Build/load `single_far_llr_background.json`.
5. Calibrate ν_{eff} from background rows in H1/L1 categories.
6. Compute LLR rank for all background and foreground single features.
7. Assign FAR through the rank-tail formula.
8. Write the single CSV with $\rho = \texttt{snglsnr[j]}$ and $y = -\log_{10} \mathrm{FAR}_{\mathrm{sd}}^{(j)}$.
9. Later merge with coherent products only after each branch has assigned calibrated FAR.

May 10, 2026 – Final Deliverable of the Single Branch

The completed single branch should produce rows like:

category	detector	ρ	χ^2	r	$-\log_{10}\mathrm{FAR}$
H1_sd	H1	`snglsnr[H1]`	`chisq[H1]`	r_{H1}	$-\log_{10}\mathrm{FAR}_{\mathrm{H1}}$
L1_sd	L1	`snglsnr[L1]`	`chisq[L1]`	r_{L1}	$-\log_{10}\mathrm{FAR}_{\mathrm{L1}}$

This is the correct endpoint for the current study: a calibrated single-detector FAR plane. The raw feature plane (ρ_m, χ^2_r) is an intermediate; the final comparable quantity is $(\rho_m, -\log_{10}\mathrm{FAR}_{\mathrm{sd}})$.